Lab 9

Question 1

```
student@Echo2: ~
student@Echo2:~$ echo "All files in my home directory" > file_report.txt
student@Echo2:~$ ls -Am ~ >> file_report.txt
student@Echo2:~$ cat file_report.txt
All files in my home directory
access.log, .bash_aliases, .bash_history, .bash_logout, .bashrc, .bl2, .cache,
cis106, .config, Desktop, Documents, .dotnet, Downloads, essentials.sh, .face,
.face.icon, file_report.txt, .fonts, .gitconfig, .gnupg, .local,
marstenhouse.sh, .mozilla, Music, participation, Pictures, .pki, .profile,
Public, Scripts, snap, sorted_access_log_by_endpoint.log,
sorted_access_log_by_uname.log, .ssh, .sudo_as_admin_successful, Templates,
.var, Videos, .viminfo, .vscode, vscode.sh, wallpapers, .wget-hsts,
wildcardpractice
student@Echo2:~$
```

Question 2

```
student@Echo2: ~
student@Echo2:~$ echo "Total number of files in my home directory" >> file_report.txt
student@Echo2:~$ ls -lA ~ | grep "^-" | wc -l
17
student@Echo2:~$ ls -lA ~ | grep "^-" | wc -l >> file_report.txt
student@Echo2:~$ echo "Total number of directories in my home directory" >> file_report.txt
student@Echo2:~$ ls -lA ~ | grep "^-" | wc -l >> file_report.txt
student@Echo2:~$ tail file_report.txt
.face.icon, file_report.txt, .fonts, .gitconfig, .gnupg, .local,
marstenhouse.sh, .mozilla, Music, participation, Pictures, .pki, .profile,
Public, Scripts, snap, sorted_access_log_by_endpoint.log,
sorted_access_log_by_uname.log, .ssh, .sudo_as_admin_successful, Templates,
.var, Videos, .viminfo, .vscode, vscode.sh, wallpapers, .wget-hsts,
wildcardpractice
Total number of files in my home directory
17
Total number of directories in my home directory
17
student@Echo2:~$
```

Question 3

student@Echo2: ~

```
student@Echo2:~$ awk -F":" '{print $1}' /etc/passwd
root
daemon
bin
sys
sync
games
man
lp
mail
news
uucp
proxy
www-data
backup
list
irc
_apt
nobody
systemd-network
dhcpcd
tss
systemd-timesync
messagebus
dnsmasq
avahi
speech-dispatcher
usbmux
cups-pk-helper
fwupd-refresh
geoclue
gnome-remote-desktop
saned
polkitd
rtkit
colord
Debian-gdm
student
_flatpak
student@Echo2:~$ awk -F":" '{print $1,$NF}' /etc/passwd
```

student@Echo2: ~

```
dhcpcd /bin/false
tss /bin/false
systemd-timesync /usr/sbin/nologin
messagebus /usr/sbin/nologin
dnsmasq /usr/sbin/nologin
avahi /usr/sbin/nologin
speech-dispatcher /bin/false
usbmux /usr/sbin/nologin
cups-pk-helper /usr/sbin/nologin
fwupd-refresh /usr/sbin/nologin
geoclue /usr/sbin/nologin
gnome-remote-desktop /usr/sbin/nologin
saned /usr/sbin/nologin
polkitd /usr/sbin/nologin
rtkit /usr/sbin/nologin
colord /usr/sbin/nologin
Debian-gdm /bin/false
student /bin/bash
_flatpak /usr/sbin/nologin
student@Echo2:~$ awk -F":" '{print $1,$NF,$6}' /etc/passwd
root /bin/bash /root
daemon /usr/sbin/nologin /usr/sbin
bin /usr/sbin/nologin /bin
sys /usr/sbin/nologin /dev
sync /bin/sync /bin
games /usr/sbin/nologin /usr/games
man /usr/sbin/nologin /var/cache/man
lp /usr/sbin/nologin /var/spool/lpd
mail /usr/sbin/nologin /var/mail
news /usr/sbin/nologin /var/spool/news
uucp /usr/sbin/nologin /var/spool/uucp
proxy /usr/sbin/nologin /bin
www-data /usr/sbin/nologin /var/www
backup /usr/sbin/nologin /var/backups
list /usr/sbin/nologin /var/list
irc /usr/sbin/nologin /run/ircd
_apt /usr/sbin/nologin /nonexistent
nobody /usr/sbin/nologin /nonexistent
systemd-network /usr/sbin/nologin /
dhcpcd /bin/false /usr/lib/dhcpcd
```

```
student@Echo2:~$ awk -F":" '{print"The user" $1,"logs in with: "$NF,"and its home dir is:" $6}' /etc/passwd
The userroot logs in with: /bin/bash and its home dir is:/root
The userdaemon logs in with: /usr/sbin/nologin and its home dir is:/usr/sbin
The userbin logs in with: /usr/sbin/nologin and its home dir is:/bin
The usersys logs in with: /usr/sbin/nologin and its home dir is:/dev
The usersync logs in with: /bin/sync and its home dir is:/bin
The usergames logs in with: /usr/sbin/nologin and its home dir is:/usr/games
The userman logs in with: /usr/sbin/nologin and its home dir is:/var/cache/man
The userlp logs in with: /usr/sbin/nologin and its home dir is:/var/spool/lpd
The usermail logs in with: /usr/sbin/nologin and its home dir is:/var/mail
The usernews logs in with: /usr/sbin/nologin and its home dir is:/var/spool/news
The useruucp logs in with: /usr/sbin/nologin and its home dir is:/var/spool/uucp
The userproxy logs in with: /usr/sbin/nologin and its home dir is:/bin
The userwww-data logs in with: /usr/sbin/nologin and its home dir is:/var/www
The userbackup logs in with: /usr/sbin/nologin and its home dir is:/var/backups
The userlist logs in with: /usr/sbin/nologin and its home dir is:/var/list
The userirc logs in with: /usr/sbin/nologin and its home dir is:/run/ircd
The user_apt logs in with: /usr/sbin/nologin and its home dir is:/nonexistent
The usernobody logs in with: /usr/sbin/nologin and its home dir is:/nonexistent
The usersystemd-network logs in with: /usr/sbin/nologin and its home dir is:/
The userdhcpcd logs in with: /bin/false and its home dir is:/usr/lib/dhcpcd
The usertss logs in with: /bin/false and its home dir is:/var/lib/tpm
The usersystemd-timesync logs in with: /usr/sbin/nologin and its home dir is:/
The usermessagebus logs in with: /usr/sbin/nologin and its home dir is:/nonexistent
The userdnsmasq logs in with: /usr/sbin/nologin and its home dir is:/var/lib/misc
The useravahi logs in with: /usr/sbin/nologin and its home dir is:/run/avahi-daemon
The userspeech-dispatcher logs in with: /bin/false and its home dir is:/run/speech-dispatcher
The userusbmux logs in with: /usr/sbin/nologin and its home dir is:/var/lib/usbmux
The usercups-pk-helper logs in with: /usr/sbin/nologin and its home dir is:/nonexistent
The userfwupd-refresh logs in with: /usr/sbin/nologin and its home dir is:/var/lib/fwupd
The usergeoclue logs in with: /usr/sbin/nologin and its home dir is:/var/lib/geoclue
The usergnome-remote-desktop logs in with: /usr/sbin/nologin and its home dir is:/var/lib/gnome-remote-desktop
```

```
www-data = /usr/sbin/nologin
backup = /usr/sbin/nologin
list = /usr/sbin/nologin
irc = /usr/sbin/nologin
_apt = /usr/sbin/nologin
nobody = /usr/sbin/nologin
systemd-network = /usr/sbin/nologin
dhcpcd = /bin/false
tss = /bin/false
systemd-timesync = /usr/sbin/nologin
messagebus = /usr/sbin/nologin
dnsmasq = /usr/sbin/nologin
avahi = /usr/sbin/nologin
speech-dispatcher = /bin/false
usbmux = /usr/sbin/nologin
cups-pk-helper = /usr/sbin/nologin
fwupd-refresh = /usr/sbin/nologin
geoclue = /usr/sbin/nologin
gnome-remote-desktop = /usr/sbin/nologin
saned = /usr/sbin/nologin
polkitd = /usr/sbin/nologin
rtkit = /usr/sbin/nologin
colord = /usr/sbin/nologin
Debian-gdm = /bin/false
student = /bin/bash
_flatpak = /usr/sbin/nologin
student@Echo2:~$ awk -F":" '{print $1" = "$NF}' /etc/passwd | grep "/bin/bash"
root = /bin/bash
student = /bin/bash
student@Echo2:~$ awk -F":" '{print $1" = "$NF}' /etc/passwd | grep "/bin/bash" >> report_users.txt
student@Echo2:~$
```

```
student@Echo2: ~
The useruucp logs in with: /usr/sbin/nologin and its home dir is:/var/spool/uucp
The userproxy logs in with: /usr/sbin/nologin and its home dir is:/bin
The userwww-data logs in with: /usr/sbin/nologin and its home dir is:/var/www
The userbackup logs in with: /usr/sbin/nologin and its home dir is:/var/backups
The userlist logs in with: /usr/sbin/nologin and its home dir is:/var/list
The userirc logs in with: /usr/sbin/nologin and its home dir is:/run/ircd
The user_apt logs in with: /usr/sbin/nologin and its home dir is:/nonexistent
The usernobody logs in with: /usr/sbin/nologin and its home dir is:/nonexistent
The usersystemd-network logs in with: /usr/sbin/nologin and its home dir is:/
The userdhcpcd logs in with: /bin/false and its home dir is:/usr/lib/dhcpcd
The usertss logs in with: /bin/false and its home dir is:/var/lib/tpm
The usersystemd-timesync logs in with: /usr/sbin/nologin and its home dir is:/
The usermessagebus logs in with: /usr/sbin/nologin and its home dir is:/nonexistent
The userdnsmasq logs in with: /usr/sbin/nologin and its home dir is:/var/lib/misc
The useravahi logs in with: /usr/sbin/nologin and its home dir is:/run/avahi-daemon
The userspeech-dispatcher logs in with: /bin/false and its home dir is:/run/speech-dispatcher
The userusbmux logs in with: /usr/sbin/nologin and its home dir is:/var/lib/usbmux
The usercups-pk-helper logs in with: /usr/sbin/nologin and its home dir is:/nonexistent
The userfwupd-refresh logs in with: /usr/sbin/nologin and its home dir is:/var/lib/fwupd
The usergeoclue logs in with: /usr/sbin/nologin and its home dir is:/var/lib/geoclue
The usergnome-remote-desktop logs in with: /usr/sbin/nologin and its home dir is:/var/lib/gnome-remote-desktop
The usersaned logs in with: /usr/sbin/nologin and its home dir is:/var/lib/saned
The userpolkitd logs in with: /usr/sbin/nologin and its home dir is:/
The userrtkit logs in with: /usr/sbin/nologin and its home dir is:/proc
The usercolord logs in with: /usr/sbin/nologin and its home dir is:/var/lib/colord
The userDebian-gdm logs in with: /bin/false and its home dir is:/var/lib/gdm3
The userstudent logs in with: /bin/bash and its home dir is:/home/student
The user_flatpak logs in with: /usr/sbin/nologin and its home dir is:/nonexistent
root = /bin/bash
student = /bin/bash
student@Echo2:~$
```

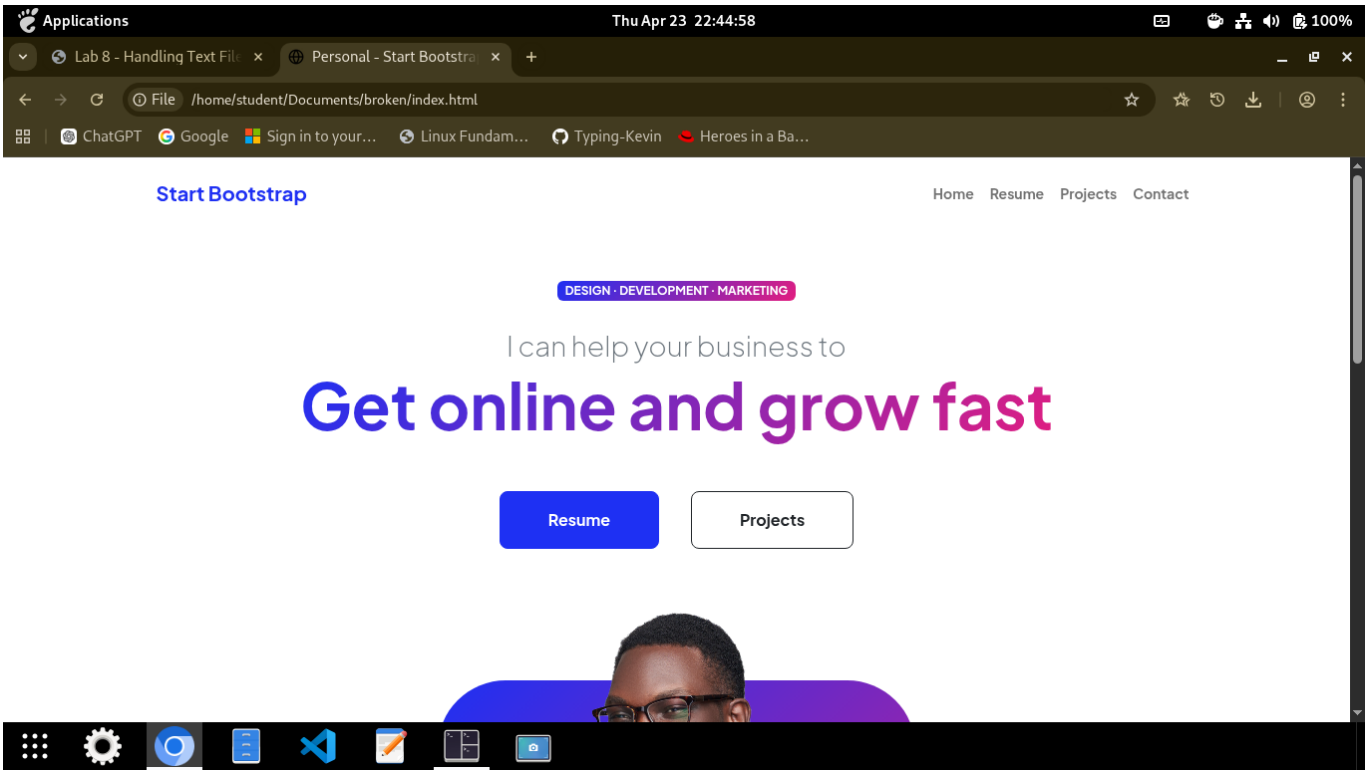
Question 4

```
student@Echo2: ~/Documents/sample_files/Csv
student@Echo2:~$ grep -i "honda" ~/Documents/sample_files/Csv/cars.csv > honda1.csv
student@Echo2:~$ head -1 ~/Documents/Csv/cars.csv > honda1.csv
head: cannot open '/home/student/Documents/Csv/cars.csv' for reading: No such file or directory
student@Echo2:~$ head -1 ~/Documents/sample_files/Csv/cars.csv > honda1.csv
student@Echo2:~$ cat honda1.csv
Car;MPG;Cylinders;Displacement;Horsepower;Weight;Acceleration;Model;Origin
student@Echo2:~$ grep -i "honda" ~/Documents/sample_files/Csv/cars.csv >> honda1.csv
student@Echo2:~$ mv honda1.csv ~/Documents/sample_files/Csv/
student@Echo2:~$ sed 's/;/,/g' honda1.csv
sed: can't read honda1.csv: No such file or directory
student@Echo2:~$ cd ~/Documents/sample_files/Csv/
student@Echo2:~/Documents/sample_files/Csv$ cd ~/Documents/sample_files/Csv/
student@Echo2:~/Documents/sample_files/Csv$ sed 's/;/,/g' honda1.csv
Car,MPG,Cylinders,Displacement,Horsepower,Weight,Acceleration,Model,Origin
Honda Civic,24.0,4,120.0,97.00,2489.,15.0,74,Japan
Honda Civic CVCC,33.0,4,91.00,53.00,1795.,17.5,75,Japan
Honda Civic,33.0,4,91.00,53.00,1795.,17.4,76,Japan
Honda Accord CVCC,31.5,4,98.00,68.00,2045.,18.5,77,Japan
Honda Civic CVCC,36.1,4,91.00,60.00,1800.,16.4,78,Japan
Honda Accord LX,29.5,4,98.00,68.00,2135.,16.6,78,Japan
Honda Civic 1500 gl,44.6,4,91.00,67.00,1850.,13.8,80,Japan
Honda Accord,32.4,4,107.0,72.00,2290.,17.0,80,Japan
Honda Civic 1300,35.1,4,81.00,60.00,1760.,16.1,81,Japan
Honda Prelude,33.7,4,107.0,75.00,2210.,14.4,81,Japan
Honda Accord,36.0,4,107.0,75.00,2205.,14.5,82,Japan
Honda Civic,38.0,4,91.00,67.00,1965.,15.0,82,Japan
Honda Civic (auto),32.0,4,91.00,67.00,1965.,15.7,82,Japan
student@Echo2:~/Documents/sample_files/Csv$ sed 's/;/,/g' honda1.csv > honda2.csv
student@Echo2:~/Documents/sample_files/Csv$
```

```
1/1                                student@Echo2: ~/Documents/sample_files/Csv
Honda_Civic CVCC,4,60.00
Honda_Accord LX,4,68.00
Honda_Civic 1500 gl,4,67.00
Honda_Accord,4,72.00
Honda_Civic 1300,4,60.00
Honda_Prelude,4,75.00
Honda_Accord,4,75.00
Honda_Civic,4,67.00
Honda_Civic (auto),4,67.00
student@Echo2:~/Documents/sample_files/Csv$ awk -F"," 'NR > 1{print $1,",",$3,",",$5}' honda2.csv | sed 's/ , /,/g' | sed 's/Honda /Hond
a_/g' >> honda3.csv
student@Echo2:~/Documents/sample_files/Csv$ batcat honda3.csv
       File: honda3.csv
   1   model, cylinder, hp
   2   Honda_Civic,4,97.00
   3   Honda_Civic CVCC,4,53.00
   4   Honda_Civic,4,53.00
   5   Honda_Accord CVCC,4,68.00
   6   Honda_Civic CVCC,4,60.00
   7   Honda_Accord LX,4,68.00
   8   Honda_Civic 1500 gl,4,67.00
   9   Honda_Accord,4,72.00
  10   Honda_Civic 1300,4,60.00
  11   Honda_Prelude,4,75.00
  12   Honda_Accord,4,75.00
  13   Honda_Civic,4,67.00
  14   Honda_Civic (auto),4,67.00
student@Echo2:~/Documents/sample_files/Csv$
```

Challenge Question

```
student@Echo2:~/Documents/broken$ ls
contact.html  favicon.ico  index.html  profile.png  projects.html  resume.html  scripts.js  styles.css
student@Echo2:~/Documents/broken$ tree -a
.
├── contact.html
├── favicon.ico
├── index.html
├── profile.png
├── projects.html
├── resume.html
├── scripts.js
└── styles.css

1 directory, 8 files
student@Echo2:~/Documents/broken$ cat index.html | grep "src"
                                <img class="profile-img" src="assets/profile.png" alt="..." />
        <script src="https://cdn.jsdelivr.net/npm/bootstrap@5.2.3/dist/js/bootstrap.bundle.min.js"></script>
        <script src="js/scripts.js"></script>
student@Echo2:~/Documents/broken$ cat index.html | grep "href"
        <link rel="icon" type="image/x-icon" href="assets/favicon.ico" />
        <link rel="preconnect" href="https://fonts.googleapis.com" />
        <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin />
        <link href="https://fonts.googleapis.com/css2?family=Plus+Jakarta+Sans:wght@100;200;300;400;500;600;700;800;900&amp;display=swap
" rel="stylesheet" />
        <link href="https://cdn.jsdelivr.net/npm/bootstrap-icons@1.8.1/font/bootstrap-icons.css" rel="stylesheet" />
        <link href="css/styles.css" rel="stylesheet" />
                <a class="navbar-brand" href="index.html"><span class="fw-bolder text-primary">Start Bootstrap</span></a>
                        <li class="nav-item"><a class="nav-link" href="index.html">Home</a></li>
                        <li class="nav-item"><a class="nav-link" href="resume.html">Resume</a></li>
                        <li class="nav-item"><a class="nav-link" href="projects.html">Projects</a></li>
                        <li class="nav-item"><a class="nav-link" href="contact.html">Contact</a></li>
                                <a class="btn btn-primary btn-lg px-5 py-3 me-sm-3 fs-6 fw-bolder" href="resume.html">Resume</a>
```

```
                            <li class="nav-item"><a class="nav-link" href="contact.html">Contact</a></li>
                                    <a class="btn btn-primary btn-lg px-5 py-3 me-sm-3 fs-6 fw-bolder" href="resume.html">Resume</a>
                                    <a class="btn btn-outline-dark btn-lg px-5 py-3 fs-6 fw-bolder" href="projects.html">Projects</a>
                                    <a class="text-gradient" href="#!"><i class="bi bi-twitter"></i></a>
                                    <a class="text-gradient" href="#!"><i class="bi bi-linkedin"></i></a>
                                    <a class="text-gradient" href="#!"><i class="bi bi-github"></i></a>
                        <a class="small" href="#!">Privacy</a>
                        <a class="small" href="#!">Terms</a>
                        <a class="small" href="#!">Contact</a>
student@Echo2:~/Documents/broken$ mkdir -p assets css js
student@Echo2:~/Documents/broken$ ls
assets  contact.html  css  favicon.ico  index.html  js  profile.png  projects.html  resume.html  scripts.js  styles.css
student@Echo2:~/Documents/broken$ mv profile.png assets/
student@Echo2:~/Documents/broken$ mv scripts.js js
student@Echo2:~/Documents/broken$ mv favicon.ico assets/
student@Echo2:~/Documents/broken$ mv styles.css css
student@Echo2:~/Documents/broken$ ls
assets  contact.html  css  index.html  js  projects.html  resume.html
student@Echo2:~/Documents/broken$ ls -l | grep "^-"
-rw-rw-r-- 1 student student  9275 Mar 25  2023 contact.html
-rw-rw-r-- 1 student student 27570 Mar 25  2023 index.html
-rw-rw-r-- 1 student student  6395 Mar 25  2023 projects.html
-rw-rw-r-- 1 student student 15143 Mar 25  2023 resume.html
student@Echo2:~/Documents/broken$ firefox index.html
^CExiting due to channel error.

student@Echo2:~/Documents/broken$ Exiting due to channel error.
^C
student@Echo2:~/Documents/broken$ chromium index.html
Opening in existing browser session.
student@Echo2:~/Documents/broken$
```